



Gaussian fuzzy membership functions with fractal belief functions and an application

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ABSTRACT

One of the contributions of this study is that an innovative method is proposed for measuring uncertainties that already exists within the fuzzy membership functions. The method is based on the definition of Gaussian fuzzy membership functions by a set of fractal belief functions. The continuous Gaussian probability density function is sampled appropriately by taking into account its fast Fourier transform and the result is a discrete sequence of values evaluated at sampling points on the membership function. The position of the sample point represents the focal elements and it gives the value of the corresponding basic probability assignment of the focal element. In this study, two uncertainty measures are utilized. The uncertainty measure in the set space that is the characteristic of the choice distribution among hypotheses due to uncertain evidence. It is generalization of the Hartley's measure, $\ln\{X\}^\#$, where $\{\bullet\}^\#$ is cardinality of the set X . This is called core entropy of focal elements. The other is uncertainty measure in probability space called belief entropy of basic probability assignments. It is based on the Shannon's entropy, $p(X)\ln p(X)$. The entropies defined in fractal belief functions are the basis for solving decision-making or classification problems. After a series of experiments and tests, it was seen that defining the membership functions with Gaussin PDF can facilitate implementing both focal elements and basic probability assignments of the fractal belief functions. We saw that our porposed method had achieved both a satisfactory and advantageous results.

1. Introduction

This study proposes an innovative approach to decision-making processes by fuzzy logic systems as explained in the followings. This study is motivated by the requirement of measuring uncertainties that inherently exist within fuzzy membership functions (FMFs) in some of our applications as seen in Fig. 1 (a). This motivation has led us to define FMFs in an innovative manner. The proposed approach is that the membership functions (MFs) are constructed in such a way that they are partitioned into a set of fractal belief functions (FBFs) as shown in Fig. 1 (b). The definition of MFs by FBFs provides us to measure uncertainties within the MFs. This form of MF gives us an opportunity to design decision-making processes by using the properties of the FBF such as entropy definitions, conductivity concept and divergence definitions. In this study, a standard Gaussian probability density function (PDF) is proposed to form the MFs. Another contribution is that Gaussian probability density function is discretized for defining fractal belief functions. The Gaussian PDF is partitioned into FBFs such that each FBF is a

rectangular region (a stripe) under the curve of the function. A normal PDF has been chosen because it has an amplitude of one and a mean value of zero. This makes it a suitable function to be used as an MF. The Gaussian PDF is a continuous function. It has to be transformed into a discrete function. The PDF is sampled based on its bandwidth. A sequence of numbers is obtained by the sampling process. These values are then passed through a zero-order holder (ZOH) to generate the FBFs. The use of a Gaussian PDF facilitates both the construction of FBFs and the corresponding FMFs. The probability values can be easily determined and implemented from the Gaussian PDF.

The definition of FMFs by FBFs has several advantages. The first advantage is the quantification of uncertainty. The other important advantage is the ability to transfer information easily from one layer to the other layers within a decision-making process. We refer to this property as the conductivity capability which denotes the capability of FBFs to interact within an environment of FBFs as defined later in this study. Another advantage of the proposed method is its ability to convert a decision to a crisp value at the final stage of the process. We claim that

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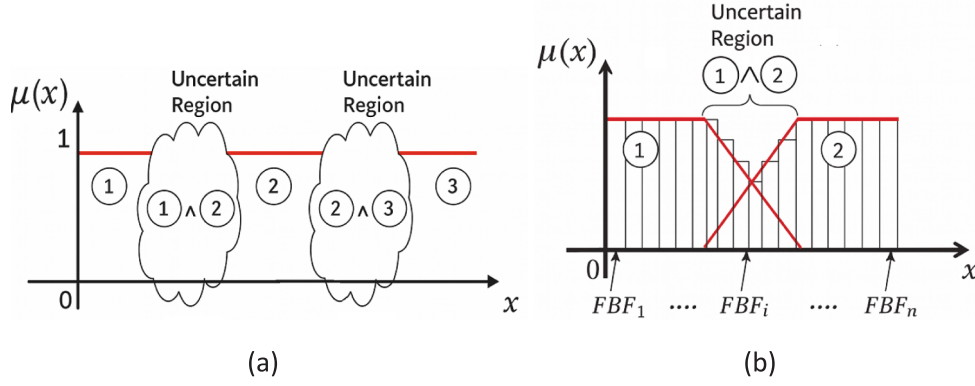


Fig. 1. (a) An MF with uncertain regions, (b) the MF described by a set of BBFs.

Table 1

Entropy Definitions for BBFs.

Symbol	Quantity	Definition ^a
h	Cardinality of a set (normalized)	$h(X_{if}) = \frac{\{X_{if}\}^\#}{\{\bigcup_{j=1}^n X_{jf}\}^\#}$
H_X	Core entropy	$H_X(FBF_F) = -\sum_{i=1}^n h(X_{if}) \ln h(X_{if})$
H_b	Belief entropy	$H_b(FBF_F) = -\sum_{i=1}^n b(X_{if}) \ln b(X_{if})$
H_W	Within set entropy	$H_W(FBF_F) = H_X(FBF_F) + H_b(FBF_F)$
sim	Similarity of FEs	$sim(X_{if}, X_{jf}) = \frac{\{X_{if} \cap X_{jf}\}^\#}{\{X_{if} \cup X_{jf}\}^\#}$
d	Focal distance	$d(X_{if}, X_{jf}) = \frac{1}{sim(X_{if}, X_{jf})} [b(X_{if}) \ln b(X_{if}) + b(X_{jf}) \ln b(X_{jf})]$
Div	Total distance	$Div(FBF_F) = \sum_{i=1}^{n-1} \sum_{j=i+1}^n d(X_{if}, X_{jf})$
H_{in}	Inner entropy	$H_{in}(FBF_F) = H_W(FBF_F) + Div(FBF_F)$
dim	Fractal dimension	$dim(FBF_F) = \frac{H_W(FBF_F)}{H_{in}(FBF_F)}$
Cap	Information Capacity	$Cap(FBF_F) = \frac{1}{H_{in}(FBF_F)}$

this formulation of FMFs can be applied to a wide range of tasks, including classification problems, decision-making approaches, and the control of dynamical systems. The concept of interaction plays a crucial role in ensuring the flow of information from one belief function to another throughout the process. For instance, in our approach, a control function based on an error function (i.e., an input evidence) must be transmitted into a dynamical system in order to generate the desired control action. In this paper, the theoretical foundations of the proposed approach are presented. The validity of the method is further demonstrated through both a simulation study and a practical application. Here a physical dynamical system is controlled by using an FBF-based fuzzy controller. The performance of the proposed controller is compared to that of a classical PID controller commonly used in the literature. The corresponding results are systematically summarized.

Since the application considered in this study specifically involves in a control problem, a comprehensive overview of the past, present, and future of fuzzy control is provided in [1]. A survey of the field can be found in [2]. One of the earliest studies on the application of fuzzy control to a simple dynamical plant was conducted by Mamdani [3]. A personal perspective by Zadeh on the evolution of system analysis and control is presented in [4]. Takagi and Sugeno introduced a fuzzy identification approach for modeling and controlling dynamical systems in their study [5]. Fuzzy logic has also been widely utilized in autonomous mobile robotics and mechanical systems [6,7]. Additionally, fuzzy logic and fractal theory have been employed for plant monitoring applications [8]. Castillo and Melin further integrated neural networks, fuzzy logic, and fractal concepts for a quality control problem [9]. Intelligent controllers operating under uncertainty using both Type-1

and Type-2 fuzzy logic were experimentally investigated in [10].

Our approach to handling uncertainty within FMFs by BBFs is grounded in the theoretical developments provided by Zadeh and then Erkmén and Stephanou in [11,16]. Further contributions involving BBFs in combination with neural network structures are reported in [13,16]. A belief-function-based divergence classifier was explored in [15]. Studies on Type-2 and interval Type-2 fuzzy logic systems have been widely reported in [16–22]. The proposed FBF-based fuzzy logic methodology in this work builds upon the fundamental contributions presented in [23–35]. For the modeling of the physical discrete control system used in our application, we have followed the frameworks and principles described in [36–38].

There are some studies related to type-2 fuzzy logic applications in the literature. A comprehensive survey on type-2 fuzzy logic is given in [39]. This paper provides a compact survey of research on type-2 fuzzy logic, highlighting its theoretical foundations, practical applications, and trends across intelligent control, pattern recognition, and classification. It reviews past and current developments, identifies emerging directions, and outlines potential future research opportunities. The main contribution is offering a clear overview of key works in the field and presenting guidance for future studies.

In the study found in [40], the authors introduce a preliminary fuzzy fractal control framework that integrates fractal dimension-based complexity analysis with fuzzy logic-based uncertainty handling to control nonlinear dynamical systems. The effectiveness of the approach illustrated through a case study on COVID-19 pandemic control. The paper in [41] tackles the long-standing problem of quantifying the total uncertainty of BPAs in Dempster-Shafer theory. Although many entropy measures have been proposed, non adequately handle both discord and non-specificity. To address this, the authors simulate the pignistic probability transformation (PPT) using a fractal-inspired framework, providing an intuitive view of how information volume changes during transformation. Based on this analysis, they introduce a new uncertainty measure, fractal-based belief entropy, which demonstrates superior performance compared to existing belief entropies in capturing total uncertainty and maintaining physical model consistency.

In the paper [42] the authors addresses the problem of fusing uncertain information originating from different theoretical frameworks by focusing on transforming fuzzy membership functions into basic belief assignments. Existing transformations based on uncertainty maximization or minimization produce extreme cases, leading to biased results. To obtain a more balanced representation, the authors propose two moderate trade-off transformations: (1) a weighted-average-based transformation, and (2) an optimization-based transformation with a weighting mechanism. The weights can be predefined or derived from prior information. Numerical and classification experiments demonstrate the rationality and effectiveness of the proposed transformations.

The study in [43] addresses the challenge of fusing highly conflicting evidence in Dempster-Shafer theory by introducing a new divergence

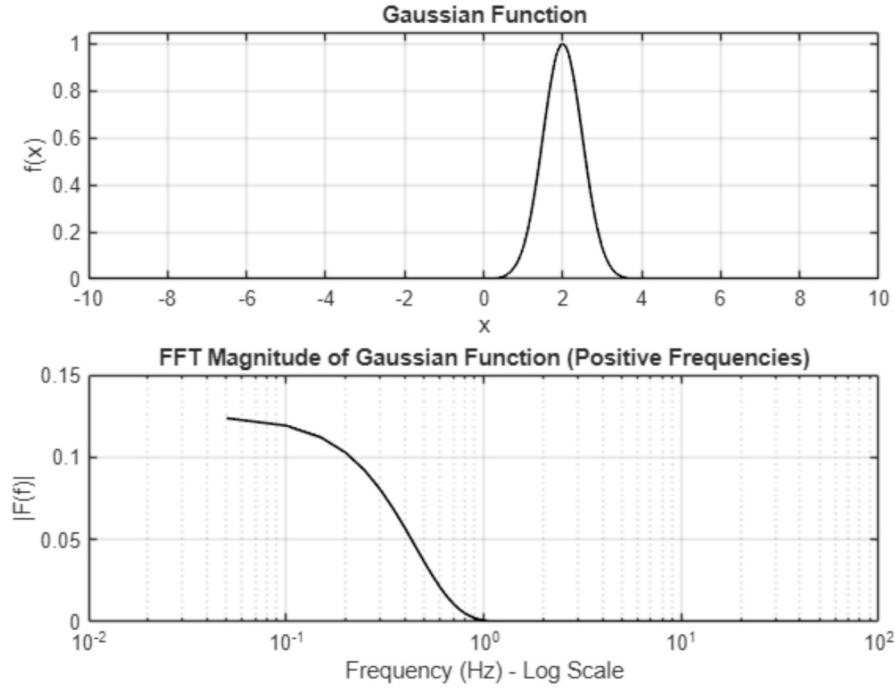


Fig. 2. The standard Gaussian PDF and its corresponding FFT are shown here. The mean is $\mu = 0$ and the standard deviation is $\sigma = 1/\sqrt{2\pi}$. The bandwidth is $BW = 3\text{Hz}$, which represents the half-power (-3 dB) frequency.

measure, the Fractal Belief Rényi (FBR) divergence, which incorporates fractal concepts into Rényi divergence to capture the time-evolving discrepancy between evidence sources. Based on this measure, the authors develop a FBReD-based weighted multisource information fusion algorithm. Experiments on real pattern-classification datasets demonstrate that the proposed method yields superior fusion and classification performance compared with existing approaches.

The article that examines uncertainty quantification in Complex Evidence Theory (CET) by analyzing the Complex Pignistic Belief Transformation (CPBT), which redistributes complex mass functions using a fractal-inspired approach. The study simulates CPBT's temporal behavior and introduces a new measure, Fractal-based Complex Belief (FCB) entropy, to evaluate uncertainty in complex basic belief assignments (CBBAs). The properties of the proposed entropy are analyzed and its effectiveness is demonstrated through numerical examples and a practical case study [44].

In our approach, we propose fractal belief functions in real domain rather than the complex domain. Although our approach mainly based on the Dempster-Shafer theory, we also define a new divergence concept that is based on the conductivity definitions [12–14] that is used for decision-making requirements.

Although our work is inspired by the studies and theories mentioned above, it adopts a distinctly different perspective. In our approach, a Gaussian PDF is selected as the basis for the FMF. A set of FBFs is used to define the FMF. The definitions of uncertainty associated with FBFs are employed, and the concept of fractal divergence is utilized to enable the flow of information among the layers of the process. To evaluate the performance of Gaussian PDF-based MFs, we compare the chosen Gaussian MF with both a classical trapezoidal MF and an exponential-type MF. As demonstrated in the subsequent sections, the basic probability assignments (BPAs) of the FBFs can be readily obtained through the discretization of the Gaussian PDF.

The paper is organized as follows. Section 2 presents the definition of problem. Section 3 describes the proposed methodology for representing FMFs using a set of FBFs. Section 4 is dedicated to the design and simulation study and the analysis of the dynamical system control. The physical implementation involving a reaction wheel used to control a

platform in a single axis is provided in Section 5. The results and discussions are given in Section 6 and concluding remarks are summarized in the final section.

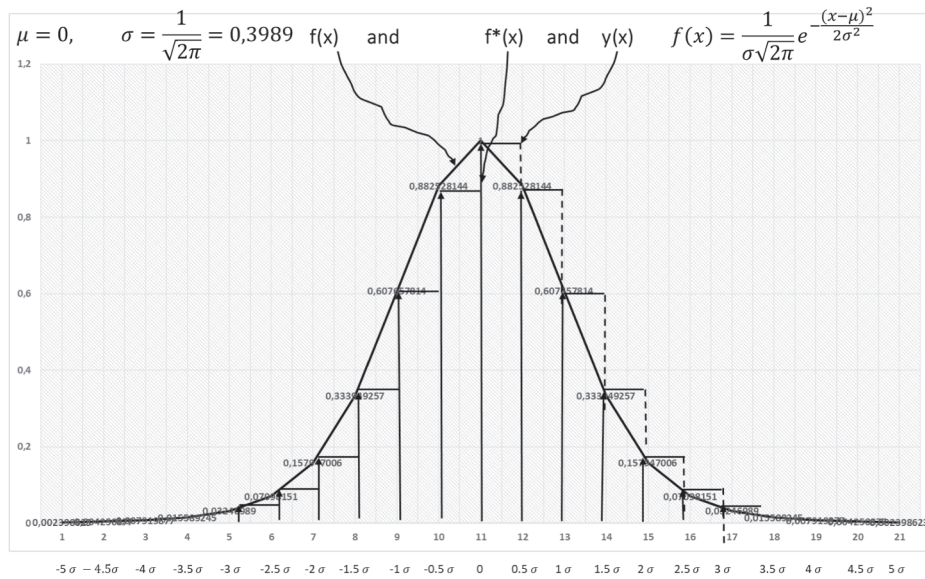
2. Problem definition

The central problem addressed in this study concerns how to quantify the uncertainties that arise when employing fuzzy logic in the control of a dynamical system. Fuzzy systems inherently contain uncertainties due to the partitions of their MFs. In practice, the boundaries of the overlapping regions within an MF are not precisely known. The degrees of uncertainty associated with these regions cannot be directly measured. Therefore, a proper framework is required to define and quantify the uncertainties present in these partitioned regions. We discuss that the FBFs provide a suitable means for measuring the uncertainties embedded in the corresponding regions of MFs. By representing an MF as a set of FBFs, it becomes possible to assess the uncertainty within these fractal structures. In this context, we propose constructing FMFs based on a Gaussian PDF. Specifically, we employ a normal PDF with mean of $\mu = 0$ and standard deviation of $\sigma = 1/\sqrt{2\pi}$. Since the amplitude of the normal PDF is one, it fits the fundamental definition of an MF. In addition, the PDF is divided into a set of FBFs by sampling the FMF. In order to determine the sampling rate, the Fourier transform is used to determine the bandwidth of the Gaussian PDF. Its well-known analytical properties make it advantageous for use in uncertainty modeling.

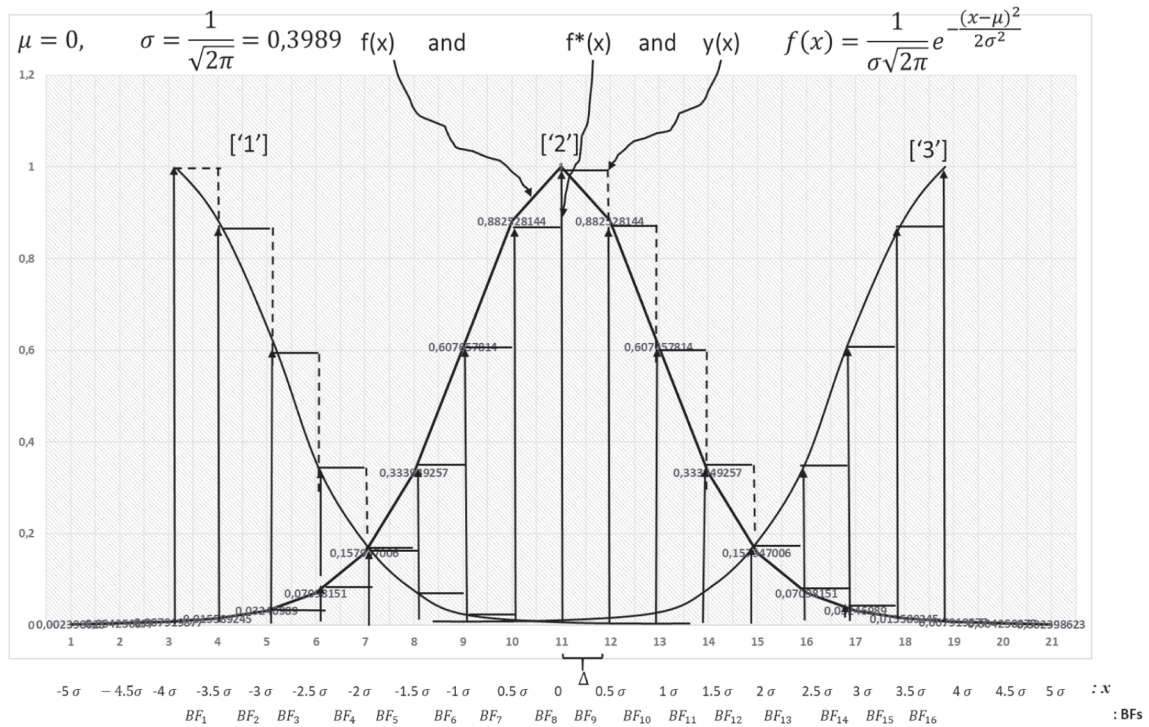
Background information on fractal belief functions is provided in the following subsections.

2.1. Fractal belief functions

An FBF consists of two main components: a set space and a probability space. Consequently, an FBF incorporates two types of uncertainties—uncertainty arising from the set space and uncertainty arising from the probability space. The formal definitions of FBFs and the associated entropy measures can be found in [13,14,17].



(a)



(b)

Fig. 3. (a) The continuous function $f(x)$, representing a standard Gaussian PDF, together with the corresponding sequence of impulse samples $f^*(x)$, and the resulting function $y(x)$ obtained after passing the impulses through the ZOH, (b) The MF composed of FBFs, where each FBF corresponds to a rectangular stripe of $y(x)$. The mean and standard deviation parameters used in the plots are provided above the figure.

The uncertainty measure in the set space that is characteristic of the choice distribution among hypotheses due to uncertain evidence. It is the generalization of the Hartley's measure, $\ln\{X\}^\#$, where $\{\bullet\}^\#$ is the cardinality of the set X . The other is the uncertainty measure in probability space. It is based on the Shannon's entropy, $p(X)\ln p(X)$. We call this measure as belief entropy of the basic probability assignments [45].

The entropies defined in fractal belief functions are the basis for solving decision-making or classification problems.

Definition 1. (Frame of Discernment, (FOD)).

The set consisting of all distinct possibilities (attributes) Y_j 's is referred to as the frame of discernment and defined as follows [13,14]:

$$FOD = \{Y_1, Y_2, \dots, Y_n\} = \{Y_j\}_{j=1}^n \quad (1)$$

Definition 2. (Singleton, (S)).

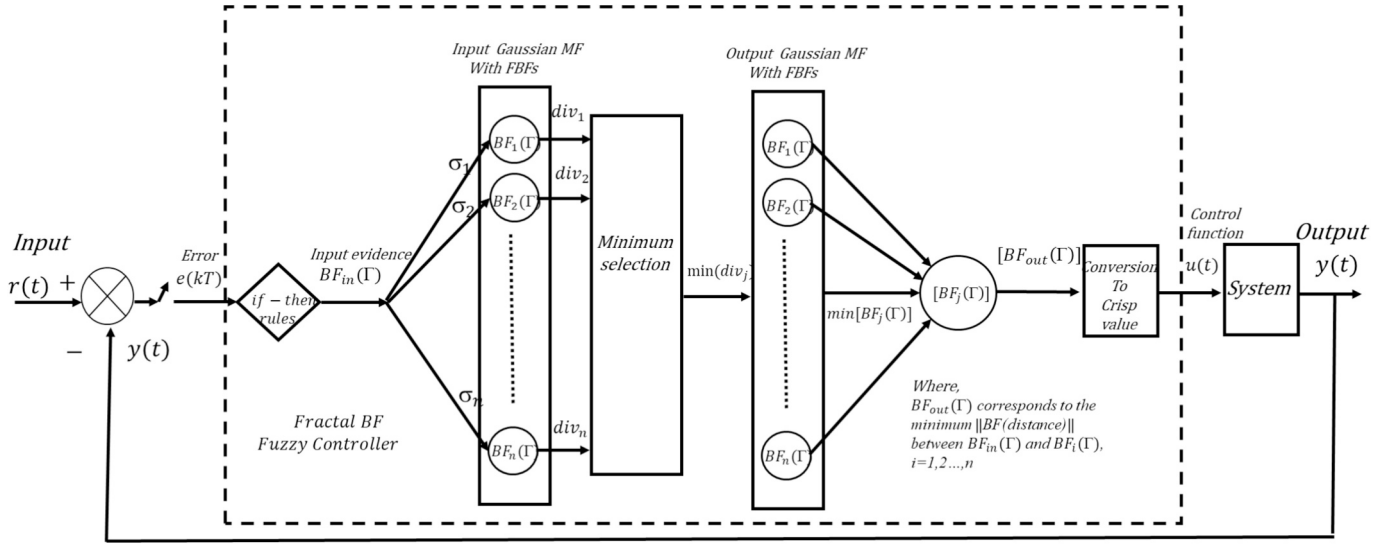


Fig. 4. The dynamical feedback system controlled by the FBF-based fuzzy controller employing the conductivity operation. The reference input $r(t)$ represents the desired rotation angle of the plant and the output $y(t) = \theta(t)$ is the actual response of the system.

Table 2
FBFs for the MF, Sampling interval $\Delta = 0.199$.

Name of Belief function	Focal Elements-FEs	Basic Probability Assignments-BPAs
FBF_1	$[[1 \setminus n]]$	$[1 \times 0.199 = 0.199]$
FBF_2	$[[1' \setminus n]]$	$[0.882 \times 0.199 = 0.175]$
FBF_3	$[[1'' \setminus n]]$	$[0.607 \times 0.199 = 0.120]$
FBF_4	$[[1' \setminus n]], [[2' \setminus n]]$	$[0.333 \times 0.199 = 0.06], [0.0709 \times 0.199 = 0.014]$
FBF_5	$[[1' \setminus n]], [[2' \setminus n]]$	$[0.157 \times 0.199 = 0.031], [0.157 \times 0.199 = 0.031]$
FBF_6	$[[1' \setminus n]], [[2' \setminus n]]$	$[0.071 \times 0.199 = 0.014], [0.333 \times 0.199 = 0.06]$
FBF_7	$[[2' \setminus n]]$	$[0.607 \times 0.199 = 0.12]$
FBF_8	$[[2'' \setminus n]]$	$[0.882 \times 0.199 = 0.175]$
FBF_9	$[[2' \setminus n]], [[3' \setminus n]]$	$[1 \times 0.199 = 0.199], [0.0]$
FBF_{10}	$[[2' \setminus n]], [[3' \setminus n]]$	$[0.882 \times 0.199 = 0.175], [0.0]$
FBF_{11}	$[[2' \setminus n]], [[3' \setminus n]]$	$[0.607 \times 0.199 = 0.12], [0.0]$
FBF_{12}	$[[2' \setminus n]], [[3' \setminus n]]$	$[0.333 \times 0.199 = 0.06], [0.071 \times 0.199 = 0.014]$
FBF_{13}	$[[2' \setminus n]], [[3' \setminus n]]$	$[0.157 \times 0.199 = 0.031], [0.157 \times 0.199 = 0.031]$
FBF_{14}	$[[2' \setminus n]], [[3' \setminus n]]$	$[0.071 \times 0.199 = 0.014], [0.333 \times 0.199 = 0.06]$
FBF_{15}	$[[3' \setminus n]]$	$[0.607 \times 0.199 = 0.12]$
FBF_{16}	$[[3'' \setminus n]]$	$[0.882 \times 0.199 = 0.175]$
FBF_{17}	$[[1' \setminus n]]$	$[1 \times 0.199 = 0.199]$

An element of a frame of discernment, Y_j ($j = 1, \dots, n$), is called a singleton.

Definition 3. (Focal Element, (FE)).

A subset of a frame of discernment is called a focal element.

$$FE = X \subseteq FOD \quad (2)$$

The entries of a focal element are constructed from the set of attributes defined in the frame of discernment as follows:

$$X = \{Y_1, Y_2, \dots, Y_m\} = \{Y_j\}_{j=1}^m, m \leq n \quad (3)$$

Definition 4. (Basic Probability Assignment, (BPA)).

The probability value assigned exclusively to the focal element $FE = X$, and to none of its other subsets, is called the BPA.

Definition 5. (Fractal Belief Function, (FBF)).

A function that maps focal elements to BPAs over a frame of discernment is called a belief function. It is represented as follows:

$$FBF = \{X_i\}_{i=1}^k; \{b(X_i)_{i=1}^k\} \quad (4)$$

where $1 \leq k \leq n$, and X_i is the i th focal element and $b(X_i)$ is the corresponding BPA.

Definition 6. (Simple Support Function, (SF)).

Each single element of FBF is called an SF.

$$SF_i = \{X_i; b(X_i)\} \quad (5)$$

where $i = 1, \dots, k, k \leq n$. In general, an FBF may contain k focal elements. The number of focal elements depends on the concepts of multi-granularity and multi-specificity of information. The multi-granularity refers to the granules of information represented within an FBF, whereas the multi-specificity relates to the composition of each focal element and the distances between different focal elements. These two characteristics quantify the uncertainties that exist both within and between the focal elements [13,14]. Such structural properties motivate the interpretation of a belief function as they give rise to a fractal dimension associated with FBFs [46].

The definitions of the uncertainty measures used in this study are provided in Table 1. These measures are required both for evaluating uncertainties and for computing similarities and distances among FBFs and divergences between the FBFs.

The definitions in Table 1 can be summarized as follows. First, the normalized cardinality of a focal element is introduced. This is followed by the definition of core entropy which measures the uncertainty contained within the focal elements. Belief entropy is then defined to quantify the uncertainty associated with the BPAs. The within-set entropy represents the total entropy measure, incorporating both core entropy and belief entropy. The similarity measure quantifies the similarity between different focal elements. The focal distance characterizes the distance between focal elements. The total distance is obtained as the sum of focal distances. The inner entropy of a belief function is formulated as the sum of the within-set entropy and the total distance of

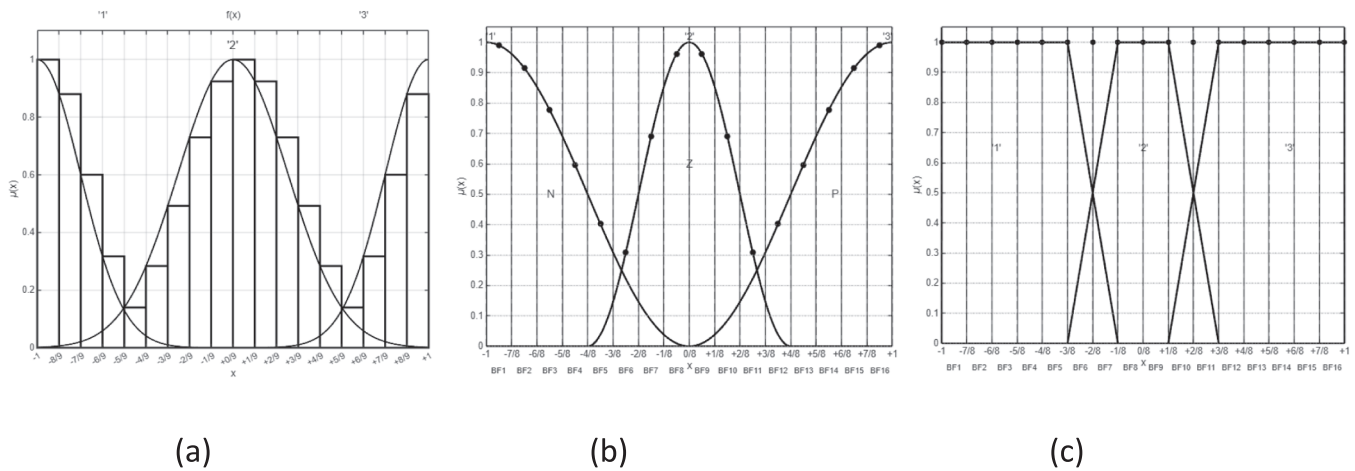


Fig. 5. Three types of MFs: (a) Gaussian-type MF, (b) Exponential-type MF, (c) Trapezoidal-type MF.

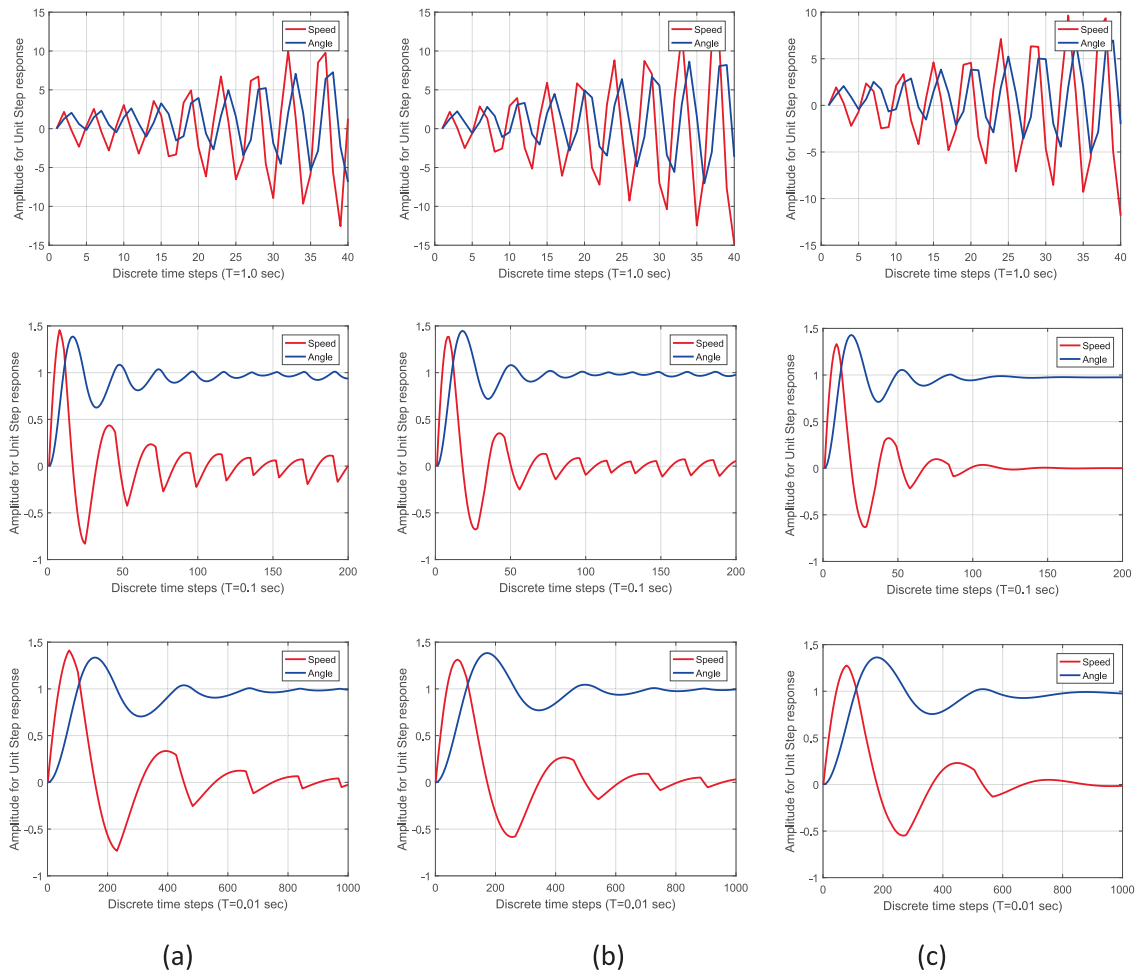


Fig. 6. Simulation results obtained using (a) Gaussian-type MF, (b) exponential-type MF, and (c) trapezoidal-type MF, shown from left to right. The plots illustrate the responses of the speed of rotation $\omega(t)$ and the angle of rotation $\theta(t)$ for each MF type employed in the study. The vertical axis shows the magnitudes of the responses and the horizontal axis gives the steps. The figures also display key dynamic characteristics, including the overshoot, rise time, steady-state error, and the stability.

the belief function. The fractal dimension of an FBF is defined as the ratio of the within-set entropy to the inner entropy. Finally, the information capacity of an FBF is defined as the inverse of the inner entropy. Sub index of BBF_F represents that the element belongs to the frame of

discernment, FOD .

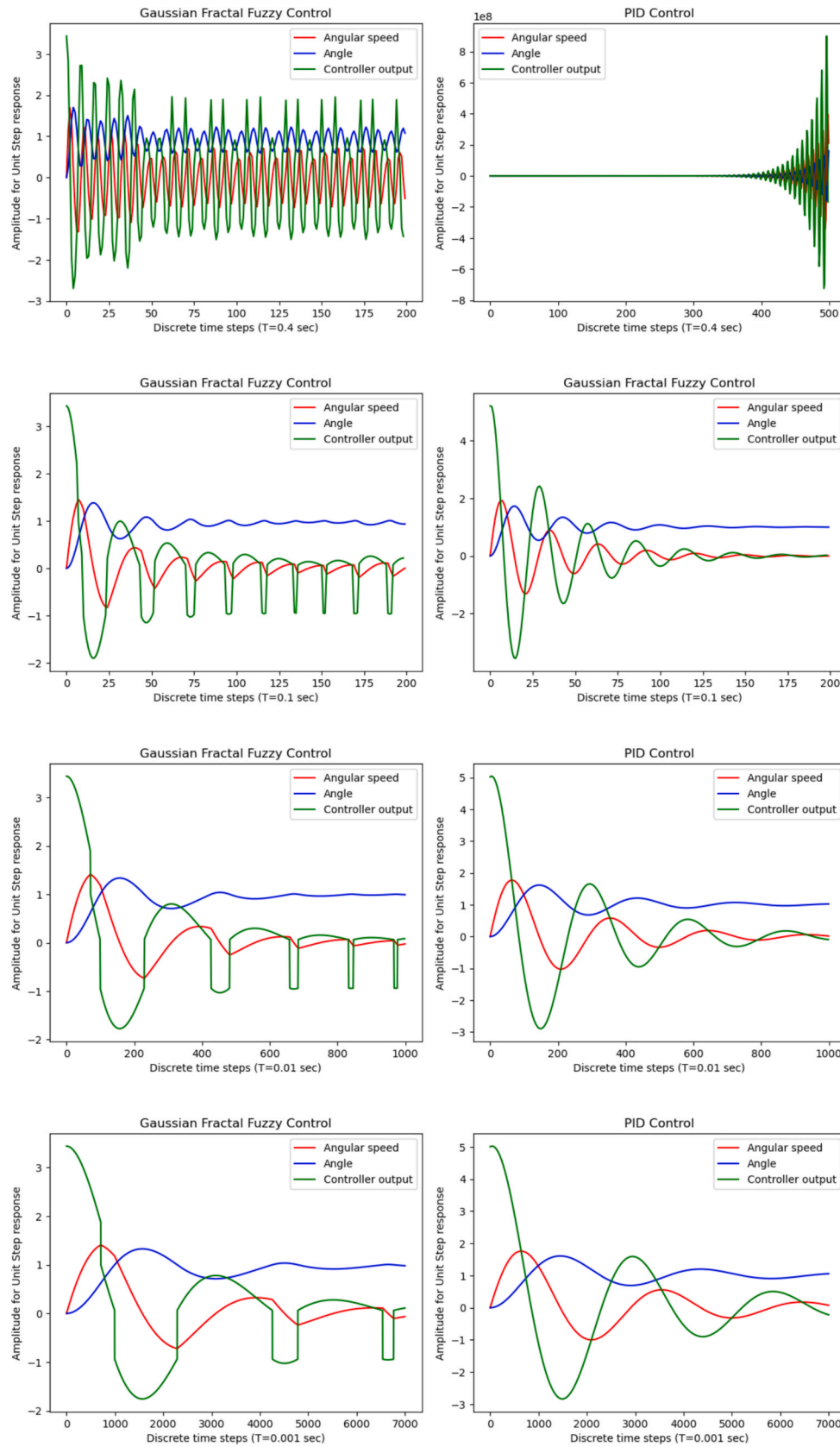


Fig. 7. The behaviors of the Fractal Gaussian Fuzzy controlled and PID controlled system are given. The plots show the responses of angular speed and the output of angle of the systems for sampling periods of $T = 0.4$ sec, 0.1 sec, 0.01 sec, and 0.001 sec. The control signals are also given.

Table 3
The test results giving the control criteria such as sampling time, maximum overshoot, rise time, settling time, and stability.

		Sampling time T (sec)	Maximum overshoot	Rise time (sec)	Settling time (sec)	Steady-state error	Stability
Gaussian Fractal Fuzzy Controller		0.4	1.703	0.8	160	Small ripples around 0.0	Stable (Highly ripples)
		0.1	1.386	1.8	12.5	0.0	Stable
		0.01	1.335	1.7	7.5	0.0	Stable
		0.001	1.327	1.6	6.5	0.0	Stable
		Sampling time T (sec)	Maximum overshoot	Rise time (sec)	Settling time (sec)	Steady-state error	Stability
PID Controller		0.4	NA*	NA*	NA*	NA*	Unstable
		0.1	1.726	1.7	14.2	0.0	Stable
		0.1	1.618	1.7	11.0	0.0	Stable
		0.001	1.608	1.6	7.5	0.0	Stable

NA*: Not applicable.

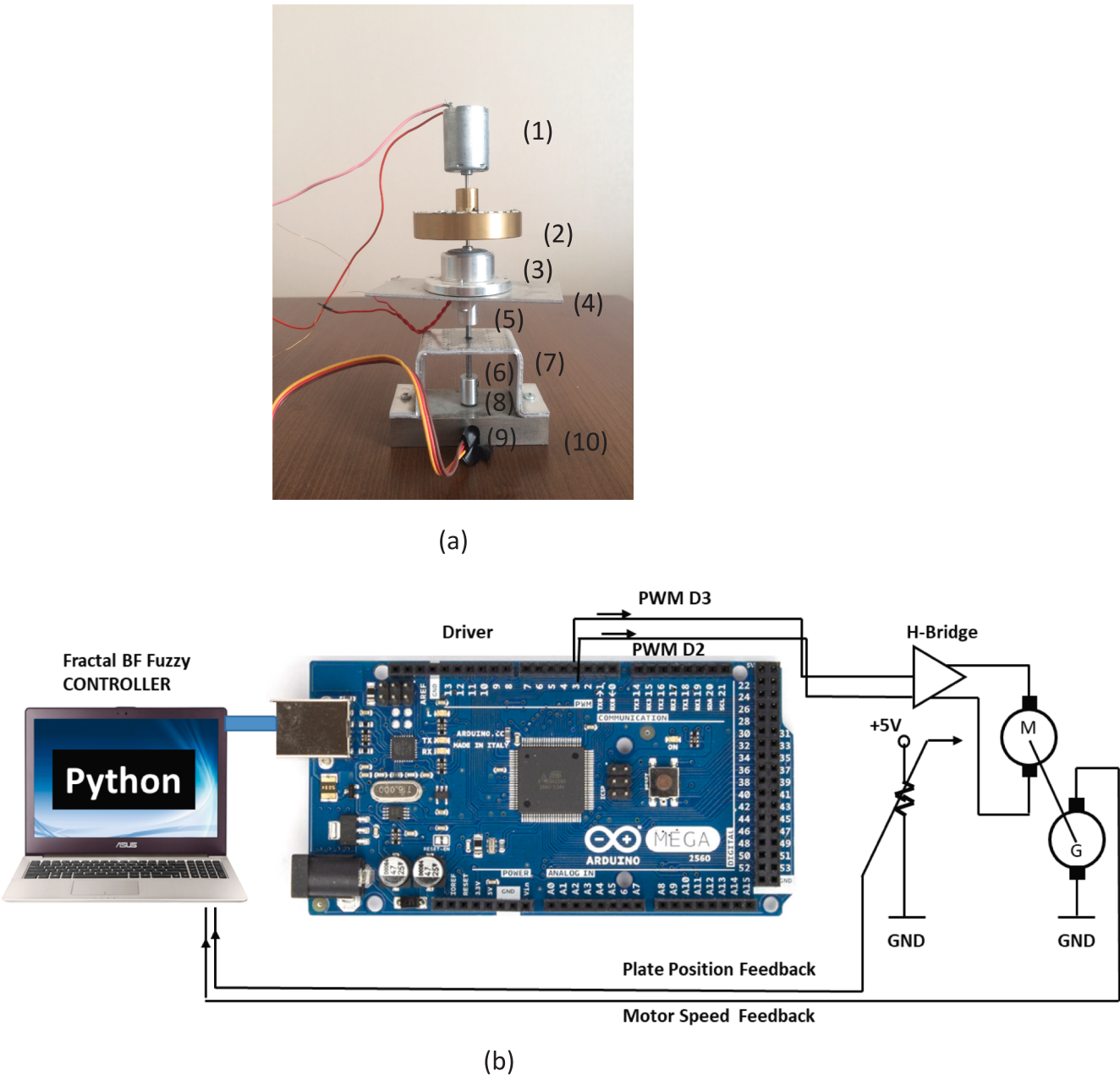


Fig. 8. (a) The testbed of the dynamical system, which is a single-axis reaction wheel system. The parts are (1) speed feedback generator, (2) brass wheel, (3) DC motor and holder, (4) platform representing one side of the CubeSat, (5) Platform shaft holder, (6) platform shaft, (7) platform holder, (8) platform shaft supporter, (9) platform rotation sensor, (10) system base, and (b) Circuit diagram of the system.

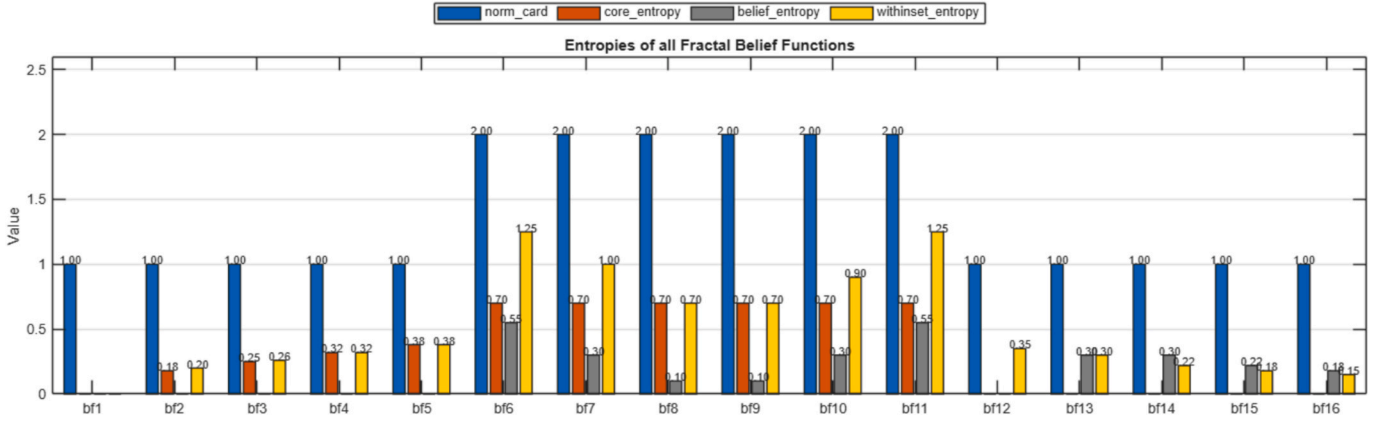


Fig. 9. Entropy values representing the measures of uncertainty present in the system. The bars in the plot correspond to the entropy-related quantities of each FBF, namely the normalized cardinality, core entropy, belief entropy, and within-set entropy.

2.2. Interaction between fractal belief functions

In this subsection, several definitions related to the interaction among FBFs are presented.

Definition 7. (Aggregate of Belief Functions).

An aggregate of belief functions is defined as a set of belief functions defined over the same frame of discernment as follows:

$$\Gamma = [FBF_{1F}, FBF_{2F}, \dots, FBF_{mF}] \quad (6)$$

where $FBF_{iF}, i = 1, 2, \dots, m$, is the fractal belief function in the frame of discernment.

The interaction between two FBFs refers to a quantitative measure of the closeness of a 'source FBF' to a 'target FBF', defined over the same frame of discernment.

Definition 8. (Directed Belief Distance).

Suppose that $FBF_{1F}, FBF_{2F} \in \Gamma$. Then the directed belief distances between FBF_{1F}, FBF_{2F} and FBF_{2F}, FBF_{1F} are defined as

$$ddis(FBF_{1F}, FBF_{2F}) = \sum_{i=1}^m \left[\left(\frac{1}{sim_{max_j}(X_{iF}, Y_{jF})} - 1 \right) + \left| b(X_{iF}) \cdot \ln \frac{b(X_{iF})}{b(Y_{max_j})} \right| \right] \quad (7a)$$

$$ddis(FBF_{2F}, FBF_{1F}) = \sum_{j=1}^n \left[\left(\frac{1}{sim_{max_i}(Y_{jF}, X_{iF})} - 1 \right) + \left| b(Y_{jF}) \cdot \ln \frac{b(Y_{jF})}{b(X_{max_i})} \right| \right] \quad (7b)$$

Notice that $ddis(FBF_{1F}, FBF_{2F}) \neq ddis(FBF_{2F}, FBF_{1F})$. Therefore, the computation of the belief distance that is defined as the divergence between FBFs requires considering both directed belief distances.

Definition 9. (Belief Divergence between the FBFs).

Suppose that $FBF_{iF}, FBF_{jF} \in \Gamma$. Then the belief divergence between two FBFs is given by

$$div(FBF_{iF}, FBF_{jF}) = ddis(FBF_{iF}, FBF_{jF}) + ddis(FBF_{jF}, FBF_{iF}) \quad (8)$$

As can be seen, an FBF may interact with any other FBF within an aggregate of belief functions. The degree of this interaction depends on the belief distance between their focal elements and their corresponding BPAs (7–8). We refer to this measure of interaction as conductivity.

Definition 10. (Conductivity).

Suppose that $FBF_{iF}, FBF_{jF} \in \Gamma$. Then, the conductivity between two FBFs is given by

$$\sigma(FBF_{iF}, FBF_{jF}) = div(FBF_{iF}, FBF_{jF}) \quad (9)$$

The application of the concept of conductivity in this process is as follows. Let a source FBF be denoted by FBF_{inF} . Let's consider an aggregate of $FBF_{jF} \in \Gamma, (j = 1, \dots, k)$. The source FBF_{inF} serves as the input to the aggregate of FBFs. FBF_{inF} conducts into each FBF in the aggregate of FBFs as follows. First, the conductivities between FBF_{inF} and each FBF_{jF} are computed individually, that is, $\sigma_j = \sigma(FBF_{inF}, FBF_{jF}), (j = 1, \dots, k)$. Next, the maximum conductivity among these values is identified. The maximum conductivity corresponds to the minimum divergence between the input FBF and the FBFs within the aggregate, and is defined as follows:

$$\sigma_{max} = \min(div(FBF_{inF}, FBF_{jF}), (j = 1, \dots, k)) \quad (10)$$

The input FBF is said to be conducted into the FBF within the aggregate that attains the maximum conductivity, σ_{max} .

A process is composed of a couple of layers of FBFs. The concept of conductivity ensures that information represented in the form of an FBF can propagate from one layer of FBFs into another FBF in a subsequent layer of FBFs within the process.

3. Methodology

The method developed in this study represents MFs of a fuzzy system by a set of FBFs, that is, by an aggregate of FBFs. The summary of the methodology is as follows. The MF employed in the system is based on the Gaussian PDF,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (11)$$

The function in (11) is adjusted to obtain a standard Gaussian PDF, that is, one with unit amplitude. The MF is then structured by partitioning this Gaussian PDF into several rectangular regions. Each of these regions is defined as an FBF, as detailed in the following subsection. Representing an MF in this manner enables the computation of uncertainties that inherently arise from the overlapping regions of MFs. Since the Gaussian PDF provides probabilistic values over the intervals of x , the FEs and BPAs can easily be determined. The corresponding BPAs are then assigned to the focal elements of the FBFs. The probability for a given interval $x \in [a, b]$ is obtained from the area under the PDF, expressed as follows:

$$bpa = Area(f(x))|_a^b \quad (12)$$

which can also be calculated by

Table 4
Entropy Values calculated during operations.

Fractal BFs	Gaussian MF H	Exponential MF H	Trapezoid MF H	Gaussian MF Hx	Exponential MF Hx	Trapezoid MF Hx	Gaussian MF Hb	Exponential MF Hb	Trapezoid MF Hb	Gaussian MF Hw	Exponential MF Hw	Trapezoid MF Hw
FBF1	1	1	1	0.00	0.00	0.00	0.32	0.00	0.00	0.32	0.00	0.00
FBF2	1	1	1	0.00	0.00	0.00	0.31	0.00	0.00	0.18	0.18	0.00
FBF3	1	1	1	0.00	0.00	0.00	0.25	0.25	0.00	0.25	0.25	0.00
FBF4	0.5, 0.5	1	1	0.69	0.00	0.00	0.23	0.33	0.00	0.92	0.33	0.00
FBF5	0.5, 0.5	1	0.5, 0.5	0.69	0.00	0.69	0.22	0.37	0.50	0.91	0.37	1.19
FBF6	0.5, 0.5	0.5, 0.5	0.5, 0.5	0.69	0.69	0.69	0.23	0.56	0.69	0.92	1.26	1.39
FBF7	1	0.5, 0.5	0.5, 0.5	0.00	0.69	0.69	0.25	0.28	0.33	0.25	0.97	1.02
FBF8	1	0.5, 0.5	1	0.00	0.69	0.00	0.31	0.06	0.00	0.31	0.75	0.00
FBF9	1	0.5, 0.5	1	0.00	0.69	0.00	0.32	0.06	0.00	0.32	0.75	0.00
FBF10	1	0.5, 0.5	0.5, 0.5	0.00	0.69	0.69	0.31	0.28	0.33	0.31	0.97	1.02
FBF11	1	0.5, 0.5	0.5, 0.5	0.00	0.69	0.69	0.25	0.56	0.69	0.25	1.26	1.39
FBF12	0.5, 0.5	1	0.5, 0.5	0.69	0.00	0.69	0.23	0.37	0.50	0.92	0.37	1.19
FBF13	0.5, 0.5	1	1	0.69	0.00	0.00	0.22	0.33	0.00	0.91	0.33	0.00
FBF14	0.5, 0.5	1	1	0.69	0.00	0.00	0.23	0.25	0.00	0.92	0.25	0.00
FBF15	1	1	1	0.00	0.00	0.00	0.25	0.18	0.00	0.25	0.18	0.00
FBF16	1	1	1	0.00	0.00	0.00	0.31	0.00	0.00	0.31	0.00	0.00
Mean												
STD												
Deviation												

Table 5

Mean Values and standard deviations for tree types of mFS.

Fractal Membership Functions	Mean value	Standard deviation
Gaussian	0.52	0.31
Exponential	0.51	0.41
Trapezoidal	0.45	0.59

$$bpa = \int_a^b f(x) dx \quad (13a)$$

The integral in (13a) takes the following form:

$$bpa = \int_{x=0}^{x+a} e^{-(x)^2} dx = \frac{1}{2} \sqrt{\pi} \text{erf}(a) \quad (13b)$$

where $\text{erf}(a)$ is the error function. However, the simulation and physical implementation of the fractal-based fuzzy system require this integration operation to be appropriately discretized. Therefore, in the following subsection, we develop the solution of the integral in discrete form.

3.1. Sampling of Gaussian probability density function

Let $f(x)$ denote the continuous Gaussian PDF given in (11). When this function is sampled appropriately, the result is a discrete sequence of values evaluated at sampling points x_n , such that $f(x_n) = f(x)|_{x=x_n}$. Assume that the sampling points are uniformly spaced, i.e., $x_n = n\Delta$, where Δ is the sampling interval, which must be chosen in accordance with the Nyquist criterion. The function in (11), together with its fast Fourier transform (FFT), is illustrated in Fig. 2. The sampling interval Δ can be readily determined from the bandwidth of the FFT plot. The bandwidth is defined as the frequency at which the magnitude $|F(f)|$ drops to 50 % of its maximum value. In this example, the bandwidth is $BW = 3\text{Hz}$, thus the sampling interval can be selected such that $\Delta \leq 1/2BW$.

After sampling, the function becomes a sequence of discrete values observed at locations x_n , and these values are given by $f(x_n) =$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_n-\mu)^2}{2\sigma^2}}. \text{ The integration of (11) yields}$$

$$p(x) = \int_0^x f(\sigma) d\sigma \quad (14)$$

The function $f(\sigma)$ is approximated by a piecewise constant function, $f_a(\sigma) = f(n\Delta)$ for $n\Delta \leq \sigma < n\Delta + \Delta$.

Accordingly, (13) is approximated as follows:

$$p(n\Delta) = \int_0^{n\Delta} f(\sigma) d\sigma \cong \sum_{l=0}^{n-1} \Delta f(l\Delta) \quad (15)$$

In fact, this corresponds to the integration operation as Δ approaches zero. Let us proceed more systematically by determining

$$p(n\Delta + \Delta) = \int_0^{n\Delta + \Delta} f(\sigma) d\sigma \cong \sum_{l=0}^n \Delta f(l\Delta) \quad (16)$$

Subtract (15) from (16),

$$p(n\Delta + \Delta) - p(n\Delta) = \Delta f(n\Delta) \quad (17)$$

In this way, only the previous value of $p(n\Delta)$ and the current value of $f(n\Delta)$ are retained, rather than all past values, in order to compute the value of the integral at $x = (n + 1)\Delta$. This yields a first-order difference equation. Accordingly, we obtain

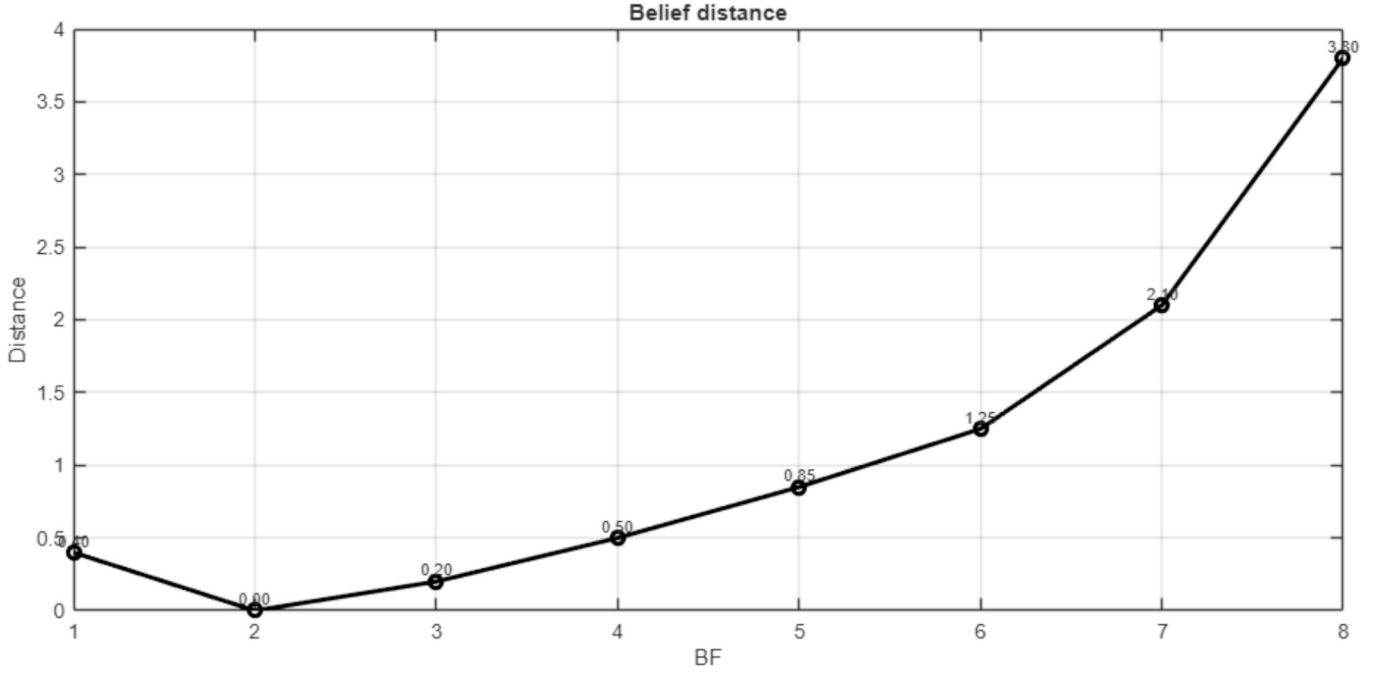


Fig. 10. Plot of the divergences. The values of divergences are calculated at each sampling instance. At each step, the minimum value of this function is converted to a crisp value for control signal that represents the output of the fuzzy controller.

$$p(m\Delta) = \Delta f(m\Delta - \Delta) + p(m\Delta - \Delta) \quad (18)$$

when $f(x)$ is approximated by a piecewise constant function, as illustrated in Fig. 2, the function $f(x)$, which corresponds to an FBF-based MF in Gaussian form, is approximated by

$$f(x) \cong \sum_{i=0}^N f_i [u(x - i\Delta) - u(x - (i+1)\Delta)] \quad (19)$$

where $u(x)$ is a unit step function that represents the interval $i\Delta \leq x < (i+1)\Delta$.

3.2. Sampling of Gaussian FBF based MF

Let us now consider the concept of impulse sampling.

Definition 10. The impulse sampler is defined as

$$f^*(x) = \sum_{n=-\infty}^{\infty} f(n\Delta) \delta(x - n\Delta) \quad (20)$$

where $\delta(x)$ is the impulse function.

When a continuous function $f(x)$ is applied to an impulse sampler, the resulting sequence of weighted impulse samples, denoted by $f^*(x)$, is shown in Fig. 3. When these weighted impulses are subsequently applied to a zero-order holder, a continuous function $y(x)$ is obtained. The impulse response of the ZOH is given by

$$g_0(x) = u(x) - u(x - \Delta) \quad (21)$$

The Laplace transform of the ZOH is

$$G_0(s) = \frac{1 - e^{-s\Delta}}{s} \quad (22)$$

If this holding action is considered as the transfer function of the ZOH, we can write it as

$$G_0(s) = \frac{Y(s)}{F^*(s)} \quad (23)$$

where

$$F^*(s) = \sum_{n=-\infty}^{\infty} f(n\Delta) e^{-ns\Delta} \quad (24)$$

and

$$y(x) = f(n\Delta), \text{ for } n\Delta \leq x < (n+1)\Delta \quad (25)$$

Hence,

$$Y(s) = \frac{1 - e^{-s\Delta}}{s} \sum_{n=0}^{\infty} f(n\Delta) e^{-ns\Delta} \quad (26)$$

The continuous functions $f(x)$ and $y(x)$ are shown in Fig. 3.

In our study, $y(x)$ is regarded as the set of fractal-dimensional FBFs that constitute the Gaussian MF. In other words, $y(x)$ can be transformed into an MF composed of a set of FBFs, expressed as

$$y(x) \rightarrow \text{MF}(\text{FBF}) = \{\text{FBF}_i\}, i = 1, \dots, n \quad (27)$$

This form of the MF provides the ability to compute the uncertainties inherent within the MF in the sense of entropy [45]. The definitions of fractal dimension [46] and entropy for the FBFs were given in Section 2. Describing an MF as a set of FBFs also enables the use of the fractal conductivity concept developed by Erkmen and Stephanou [13,14]. This concept establishes a mechanism for conducting the information from one FBF, acting as the source, to another FBF, acting as the target. Through this mechanism, the information can be transferred from one layer to another within a system composed of FBFs. Information transfer via conductivity provides additional capabilities, including classification, decision-making, and control. One important application of this approach, as discussed in this paper, is the control of a dynamical system.

3.3. Assignment of focal elements and basic probabilities to fractal belief functions

In this section, the FBF is described. The FBF is based on the standard Gaussian PDF. It has a mean value of $\mu = 0$ and a standard deviation of $\sigma = 1/\sqrt{2\pi}$. Under these parameters, the Gaussian function $f(x)$ attains a

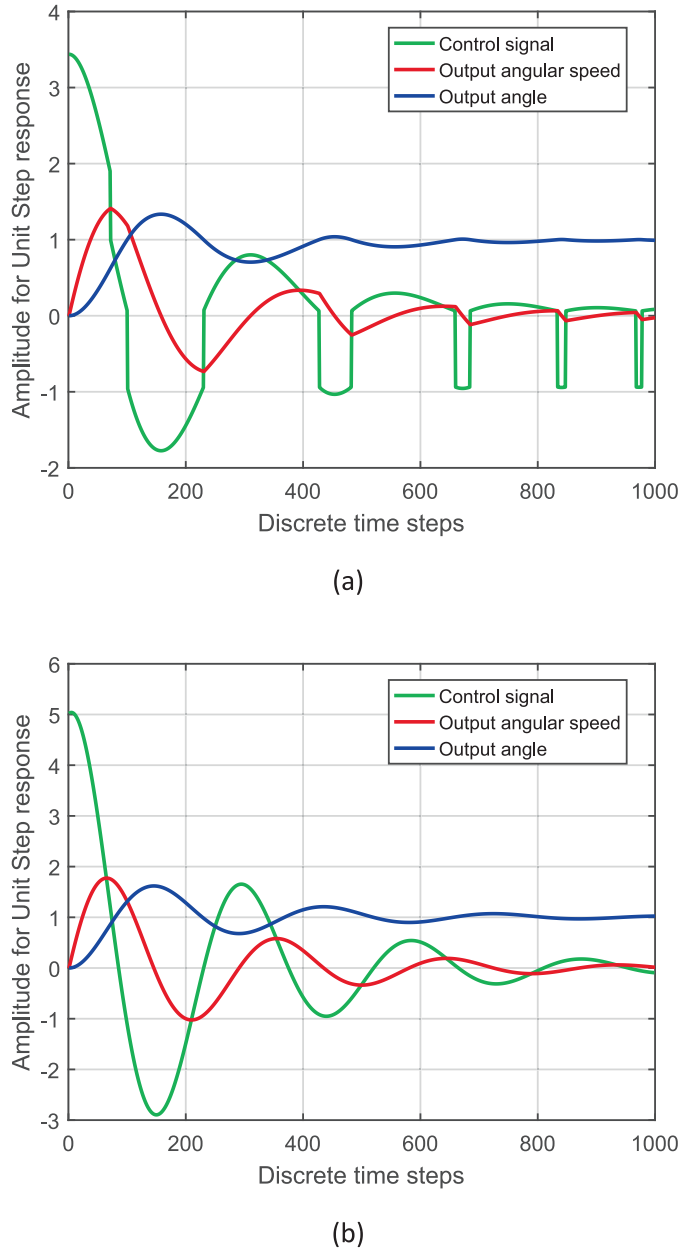


Fig. 11. System responses obtained for a sampling period of $T = 0.01$ s: (a) response under the fractal FBF fuzzy controller, and (b) response under the classical PID controller.

magnitude of one at its mean value. The function $y(x)$, obtained by sampling of $f(x)$ and passing it through the ZOH, is a collection of rectangular segments. Each of these rectangles corresponds to a region that defines an individual FBF. In this manner, the resulting MF consists of three primary regions, denoted as 'N', 'Z', and 'P', encoded as '1', '2', and '3', respectively. These codes represent the focal elements $[1]$, $[2]$, or $[3]$, or their combinations such as $[1'$ and $2']$ or $[2'$ and $3']$, etc.

The BPAs of the FBFs correspond to the areas of the rectangular segments in the function $y(x)$. In the uncertain (overlapping) regions of the MF, each region contains two rectangular segments: one belonging to a given part and the other to its neighboring part. Consequently, an FBF in such a region may have one or two focal elements, with a BPA assigned to each FE. For example, the 9th FBF (FBF_9) contains a single rectangular region, whereas the 6th FBF (FBF_6) contains two rectangular regions in Fig. 3. The focal element associated with FBF_9 is $[2 \setminus n]$ and its corresponding BPA is $.bpa_{1,9} = f(x = 0.5\sigma) \bullet \Delta = 1 \bullet 0.5 \bullet \frac{1}{\sqrt{2\pi}} = 0.5 \bullet$

$$0.398 = 0.199$$

where the subscripts indicate that the first value corresponds to the BPA of the first focal element, and the second value corresponds to the BPA of the second focal element.

The focal elements of the 6th FBF are $[1 \setminus n]$ and $[2 \setminus n]$. The BPAs that are corresponding to the focal elements are

$$bpa_{1,6} = f(x = 1.5\sigma) \bullet \Delta = 0.333 \bullet 0.5 \bullet \frac{1}{\sqrt{2\pi}} = 0.066$$

$$bpa_{2,6} = f(x = 1.5\sigma) \bullet \Delta = 0.0709 \bullet 0.5 \bullet \frac{1}{\sqrt{2\pi}} = 0.014$$

Hence, the FBFs are

$$BF_6 = [[1 \setminus n], [2 \setminus n]; [0.066, 0.014]],$$

$$BF_9 = [[2 \setminus n]; [0.199]].$$

4. Design and simulation

We simulate the dynamical system shown in Fig. 4. The system is a second-order dynamical system. Its discrete state equation and response equation are given by

$$\begin{bmatrix} x_1[(k+1)T] \\ x_2[(k+1)T] \end{bmatrix} = \begin{bmatrix} e^{-T} & 0 \\ -e^{-T} & 1 \end{bmatrix} \begin{bmatrix} x_1(kT) \\ x_2(kT) \end{bmatrix} + \begin{bmatrix} 1 - e^{-T} \\ T - 1 + e^{-T} \end{bmatrix} u(kT) \quad (28)$$

$$c(kT) = x_2(kT) = R = \theta_R(kT)$$

$$\dot{c}(kT) = x_1(kT) = \omega(kT) = 0$$

In this system, which is a rotational dynamical system, the state variable x_1 represents the angular speed, and the state variable x_2 is the angular position. Therefore, x_1 is interpreted as the rotational speed, and x_2 as the rotation angle. The system is controlled such that the output would be a constant angle.

4.1. Controller design

The controller is designed as an FBF-based fuzzy controller in which the MFs of both the error signal and the control signal are represented by the FBFs. The frame of discernment over which the FBFs are defined represents all possible attributes and is given by

$$FOD = \{Y_j\}_{j=1}^3 = \{Y_1, Y_2, Y_3\} = \{N, Z, P\} = \{1', 2', 3'\} \quad (29)$$

The frame of discernment contains three focal elements: N representing 'Negative', Z representing 'Zero', and P representing 'Positive'. Since these elements are alphabetic characters, we encode them numerically as '1', '2', and '3', respectively. The corresponding BPAs are obtained by the areas of the sampled Gaussian PDF, $y(x)$. The FBFs that constitute the MF, along with their focal elements and BPAs, are presented in Table 2.

4.1.1. The algorithm of the process

The algorithm for controlling of the dynamical system is as follows.

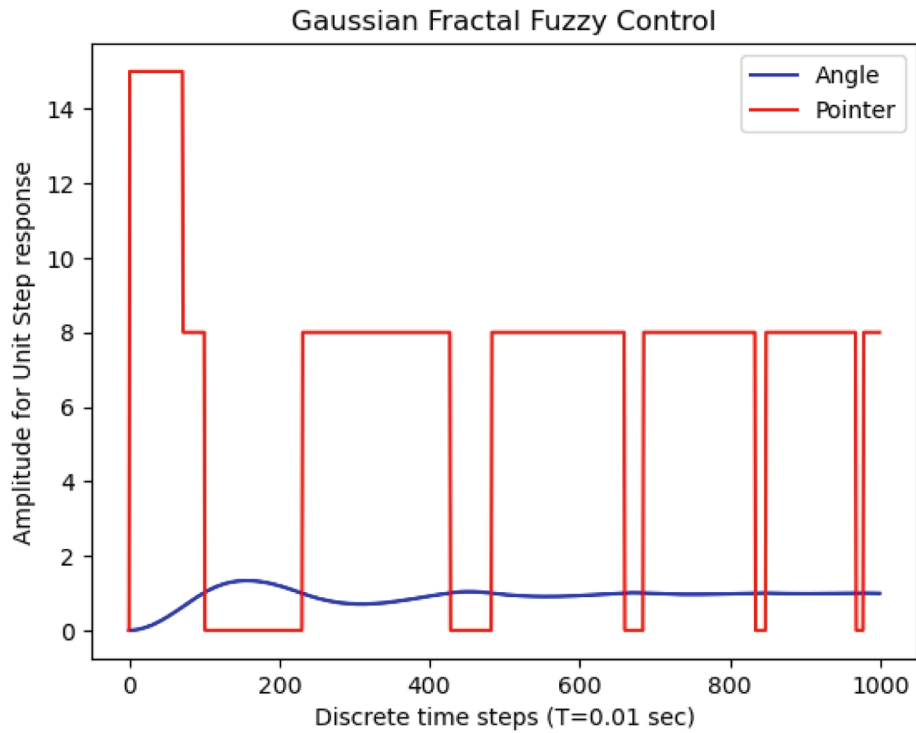
Step 1. Read the files containing the error and control MFs, which are stored as text files on the hard disk. Additional information, such as core entropy, belief entropy, within-set entropy, focal divergence, inner entropy, capacity, dimension, frame entropy, and the number of focal elements, is also retrieved from these files.

Step 2. Define the variables associated with the error input and controller output.

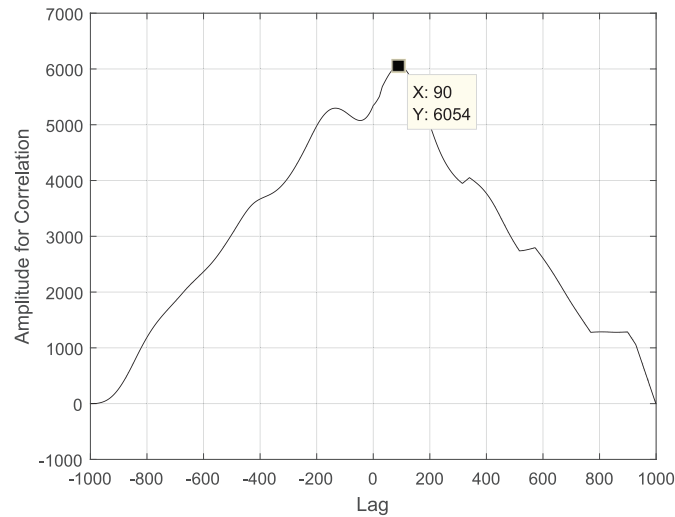
Step 3. The state equation of the system.

Step 4. Obtain the error, $e(t) = r(t) - y(t) = \theta_r(t) - \theta_y(t)$, from the feedback.

Step 5. Convert the error signal $e(t)$ into the input evidence, which is FBF_{in} .



(a)



(b)

Fig. 12. (a) Plots of the system response and the values of the pointers of minimum divergences at each sampling instances; (b) Cross-correlation function for the pointer values of the minimum divergences and the response of the system. The maximum point of the cross-correlation function is about the settling time of the system response.

Step 6. Calculate entropies of the error FBF_{in} .

Step 7. Calculate the belief divergences between the error FBF_{in} and each FBF_i , ($i = 1, \dots, n$), within the aggregate Γ of the input MF.

Step 8. Make a list of divergences calculated at Step 7.

Step 9. Find the minimum divergence within the list of divergences. Use this minimum value as the maximum conductivity term and record a pointer indicating its position in the list.

Step 10. Use the pointer to select the corresponding FBF_i from the set Γ , and assign this belief function as the controller output FBF_{out} (Fig. 4).

Step 11. Convert the control information FBF_{out} to a crisp value using the output MF.

Step 12. Apply the crisp value of the controller output $u(t)$ to the system input.

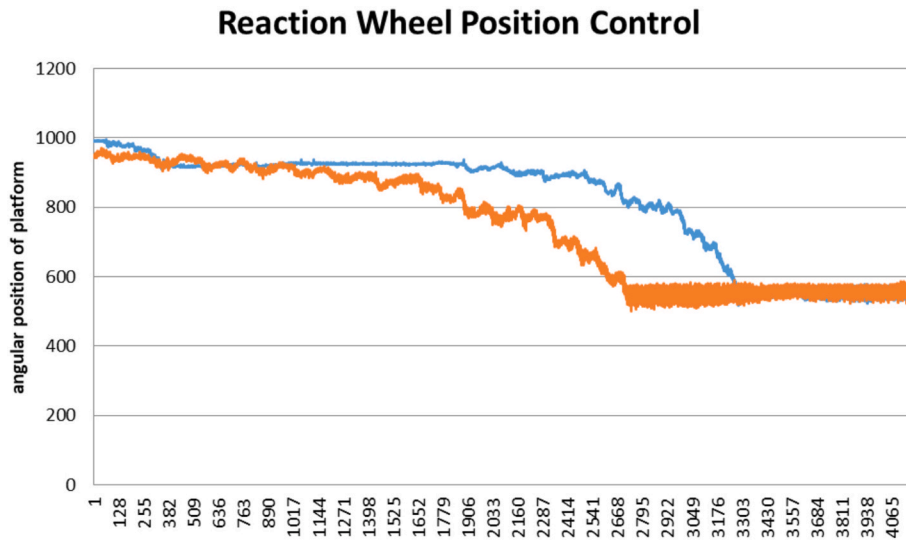


Fig. 13. In this setup, the system output corresponds to the rotation angle of the platform. The responses of the actual system are shown, where the vertical axis represents the angular position and the horizontal axis represents the sampling instances. Two responses are displayed: the first, reaching steady state at $k = 2668$, corresponds to a sampling period of $T = 0.001s$ (orange curve), while the second, reaching steady state at $k = 3303$, corresponds to a sampling period of $T = 0.0005s$ (blue curve).

Step 13. Obtain the system output, $y(t) = \theta_y(t)$.

Step 14. Return to Step 4 and repeat the procedure.

4.2. Simulation and tests with three different methods

The simulations were carried out using three different types of MFs that are the Gaussian MF, the exponential MF, and the trapezoidal MF (Fig. 5).

The Gaussian-type MF, the exponential-type MF, and the trapezoidal-type MF are based on the following functions, respectively: $f_G(x) = e^{-ax^2}$, $f_E(x) = e^{-bx} \sin x$, and $f_T(x) = cx$. The results of the responses are demonstrated in Fig. 6. The response is the rotation angle, $y(t) = \theta_y(t)$. The dynamical system is a two-dimensional system. The state variables are the angle of rotation and the speed of rotation, $x_1(t) = \theta_y(t)$ and $x_2(t) = \omega_y(t)$, respectively. When the output $y(t) = \theta_y(t)$ reaches the desired reference value, the rotational speed decreases to zero, as seen in Fig. 6. The performance of the proposed method is compared to that of controllers employing exponential and trapezoidal MFs.

4.2.1. The responses of the system for three kinds of membership functions

The Gaussian-type MF satisfies the requirements of the fuzzy controller nearly as effectively as the exponential and trapezoidal MFs. As expected, increasing the sampling period (e.g., to 1 s) results in an unstable response, as shown in Fig. 6(a). Conversely, reducing the sampling period stabilizes the system response. When the sampling period decreases, the rise time decreases, the overshoot and undershoot become smaller, and the steady state error is settled down to zero. This type of response is expected.

The control of a system by using the fractal fuzzy membership functions is the same as the well-known fuzzy control approach. The system is a sampled data system or a discrete-time system. The stability for such a system is mainly based on the selection of the sampling period, T in seconds. The responses of the system for different values of T are given here. As the value of T increases, the system begins to be unstable. As the value of T is decreased, the system becomes more stable and oscillations become smaller. The rise time and settling times decrease.

4.2.2. Comparison of Gaussian fractal fuzzy and PID controllers

There are a couple of control applications in the literature [39–44]. However, we discuss the responses of the Gaussian fractal fuzzy

controller that is compared to the responses of the PID controller. The reason for it is due to the fact that the PID control is very common and widely used in the control applications.

The state equation of the PID controller is given by the following expressions:

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} e(k) \quad (30)$$

$$u(k) = \left[T \quad -\frac{1}{T} \right] \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + (K_p + K_i \cdot T + K_d \cdot \frac{1}{T}) e(k) \quad (31)$$

where $x_1(k)$ is the integral state variable, $x_2(k)$ is the derivative state variable, and $e(k)$ is the feedback error of the system. The responses of both controllers are given in Fig. 7.

The test results of the system controlled by the Gaussian fractal fuzzy controller and PID controller are given in Table 3.

The systems are unstable for $T > 0.4sec$. The control criteria can be improved when the sampling time decreases. As you see, when the sampling time is selected smaller than this value, the maximum overshoot, rise time, and settling time decreases. The ripples on the responses become smaller and the systems become stable. Both of the control methods have similar results on the control criteria, although the system with PID controller becomes unstable for $T \geq 0.4sec$. These responses were expected because of the following facts. The PID controller includes an integral control action which has a transfer function as $T(s) = 1/s$. The integral term has a zero (or an eigenvalue of system matrix) at the origin of the complex plane. Similarly, the inertia of the reaction wheel connected to the shaft of the electrical motor and the platform have a zero at the origin of the complex plane. Therefore, the zero at the origin results in a zero steady state error.

Our system is a second-order system. Its characteristic polynomial is given as $(s^2 + 2\zeta\omega_n s + \omega_n^2)$. The maximum overshoot is affected by the coefficient of the first-degree term $2\zeta\omega_n$ of the characteristic polynomial of the second-order system. This coefficient appears at the characteristic polynomial because of the availability of frictional and resistive parameters within the system. The gain of the derivative controller K_D also appears in this coefficient. If this coefficient is large enough, then the overshoot becomes smaller. In our actual system, the frictional and resistive parameters are available more than enough, a desired response is obtained only by decreasing the value of the sampling period T .

5. Physical system

The physical process is as follows. The system is a dynamical testbed that operates as a reaction wheel mechanism rotating about a single axis. It consists of a DC motor that drives a wheel mounted on its shaft, along with a metallic square platform positioned on the same rotational axis that is fixed to a separate shaft. A potentiometer is mounted on the shaft of the platform in order to get feedback information for the rotation angle. A DC generator is coupled to the motor shaft to provide feedback information on the motor's angular speed. The reaction wheel system is designed and manufactured to test the proposed method.

The hardware implementation of the system is shown in Fig. 8. It consists of a PC-based controller, an Arduino Mega processor with its driving circuitry, a motor, a generator, and a platform equipped with a potentiometer. The operation of the system is as follows. The FBF fuzzy controller and entropy-related calculations are executed on a PC using Python 3.8. The motor is driven by an Arduino Mega processor and a driver circuitry. Feedback information for rotational angle is obtained by a potentiometer. The angular speed is fed back by a DC generator.

6. Results and discussion

The results obtained from both the simulation and the operation of the real system are presented as follows. Fig. 9 illustrates the entropy values, which quantify the inherent uncertainties present in the MFs in fuzzy controller.

In Table 4, the entropy values are given only for the uncertain regions of the MF. In the certain regions of the MF, both the core entropies and the belief entropies are zero.

An FBF consists of a set space and a probability space, and it possesses two types of entropy measure so called the core entropy, H_X , and the belief entropy, H_b . The total entropy of an FBF, referred to as the within-set entropy, H_w , is defined as the sum of these two components, i. e., $H_w = H_X + H_b$.

Table 4 presents all entropy values associated with three types of MFs considered in this study. Since each MF consists of n FBFs (with $n = 16$ in this study), the table includes the entropy measures for all of them. The mean values of the total entropies (i.e., within-set entropies H_w) and their corresponding standard deviations for the three types of MFs are summarized as follows (Table 5).

The entropy values are the basis for calculating divergences. The plot of the belief divergence values is given in Fig. 10. The minimum divergence can be identified from this plot.

The control signal together with the output angular speed and the output angle are shown for both fractal fuzzy controller and the PID controller in Fig. 11. As shown, the control signal exhibits discontinuities at the transition instants associated with the local maxima and minima of the output signals. Since the output angle is the integral of the angular speed, the plot of it is a smooth signal. It also converges to a steady-state value of zero. Hence, the desired output angle is obtained with zero steady-state error.

The studies resembling our approach are given in [39–44]. The paper in [39] is an overview of the works highlights the theoretical foundations on type-2 fuzzy logic. It is a comprehensive study in intelligent control. Since the results on this area are provided concisely, we prefer here to compare our results to the classical and widely used PID controller. The comparison results show that our approach based on the FBF based FMFs are satisfactory.

The correlation between entropy measurements and control performance is carried out as follows. The divergence values calculated at each step for selecting control value at this instance. A pointer is used to determine the position of the FBF within the output FMF. The pointer that points the position of the minimum divergence is used to select the crisp value of the FBF. The value of the pointer is the position of the FBF within the FMF. The pointer values therefore are correlated to the controller signal outputs and hence it is correlated to the output

response of the system. In this analysis, the divergence values at each sampling instant and the values of the system's responses are used. The conductivity concept is used to obtain the interactions of FBFs. The conductivity is defined as the divergence between FBFs. The values of divergences between the input FBF and the output FBFs of the output membership function and the pointer of the minimum divergence are needed for obtaining crisp values of the controller output. The value of divergence and the pointer of minimum divergence are calculated based on the core and belief entropies. The value of the pointer of the minimum divergence among the output FBFs is correlated to the response of the system. Therefore, a cross-correlation between the pointer values and the system response at each sampling instant has to be obtained. The plots of the pointers of minimum divergences and the system response are used to find the cross-correlation between them. The cross-correlation function is plotted as shown in Fig. 12. As seen, the maximum value of the cross-correlation function is about the settling time. This provides that the entropy values can be used for making proper decision among the output FBFs and obtaining a desired response for a dynamical system.

The plots presented in Fig. 13 illustrate representative responses of the actual system. In these experiments, small values of the sampling period T were selected. Under these conditions, no overshoot is observed, and the system exhibits a smooth response.

The platform's angular position begins from the rightmost value. As shown, the platform rotates from an initial angle of 1.0 to a final angle of 0.6. For example, this corresponds to a rotation from 90° to 54°. The noise observed at steady state is attributable to mechanical vibrations in the testbed, even though the steady-state angular speed is zero.

7. Conclusion

In this study, FBFs are employed to define MFs. This approach offers several advantages. First, it enables the quantification of uncertainties that naturally arise in the uncertain regions of MFs. Second, it provides an appropriate resolution for implementing MFs in practical systems. Third, it facilitates the transformation of logical outcomes into crisp values in a straightforward manner.

The primary motivation of this work was to measure the uncertainties present within the uncertain regions of MFs. Building upon this requirement, we developed FBF-based MFs. Since constructing FBFs requires mapping focal elements to their corresponding BPAs, Gaussian-type MFs were selected due to their convenient probabilistic structure. To evaluate this approach, the performance of Gaussian-type MFs was compared to that of trapezoidal and exponential MFs. As demonstrated in the preceding discussions, FBF-based MFs exhibit performance comparable to these alternative MF types. Additionally, the degrees of MF at the sampling instances for Gaussian-type MFs can be obtained directly from the Gaussian PDF, eliminating the need for additional computations. This property significantly simplifies the implementation of fuzzy control applications.

The proposed method has been applied to the control of a reaction wheel driven by a DC motor, and the results have proven highly satisfactory. The approach can be implemented on a physical system with ease, demonstrating its practical applicability. These promising outcomes motivate the extension of the method to other domains, such as measurement, classification, and decision-making processes.

The degree of uncertainty can be numerically quantified through the use of FBF-based MFs. In this study, the responses of the system controlled by three different MF types, as well as by a PID controller, have been examined. As known, the PID controller operates in discrete time, using signals sampled every T seconds. In contrast, a fuzzy controller employs MFs defined by FBFs, which are typically continuous functions of an independent variable x . This variable is sampled based on the bandwidth of the membership function. The degree of membership ranges between zero and one, and the specific shape of an MF depends on the application.

In this study, Gaussian PDFs were chosen to represent the MFs. A normal PDF was selected because it provides a convenient structure for MF representation so that the magnitude of the PDF is one and its mean value is zero. The PDF is sampled according to its bandwidth, generating a sequence of discrete values. These values are then passed through a zero-order holder, from which the fractal belief functions are formed. A total of n FBFs is obtained, and the output of the fuzzy controller is one of these FBFs. This FBF is subsequently converted into a crisp value to drive the continuous dynamical system.

Since the output of the fuzzy controller is a discrete-time signal, pulses with varying widths appear in the output. The sampling rate of the FBFs can be increased when higher resolution in the fractal fuzzy MF is required. As seen, the system responses under fuzzy control are satisfactory: the steady-state error is zero, and the behavior is consistent with the average value of the applied pulse widths.

As demonstrated, the rotation angles converge to their desired steady-state values with zero steady-state error, and the angular speeds similarly approach zero radians per second. Although minor mechanical vibrations are present in the physical system, the results confirm that the proposed Gaussian FBF-based fuzzy control approach can be effectively and reliably applied to the control of physical processes.

For future work, we plan to apply the proposed method to various problems, including radar signal detection and classification, as well as attitude control of satellite systems. We assert that the proposed approach is well suited for applications requiring intelligent capabilities in detection, estimation, classification, and decision-making processes.

CRedit authorship contribution statement

Sedat Nazlibilek: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization.
Deniz Karacor: Visualization, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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